

THANGOOL, Australia

WMO: 943760

Lat: 24.4936S

Lon: 150.5708E

Elev: 194

StdP: 99.02

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.9	3.5	-6.5	2.2	15.5	-4.0	2.8	13.1	7.8	19.9	7.1	19.9	1.2	100	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	35.9	21.6	34.4	21.5	33.2	21.3	24.7	30.4	24.1	29.8	23.5	29.3	3.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.5	26.7	22.6	17.7	26.5	21.9	16.9	26.4	76.1	30.6	73.5	29.7	71.3	29.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	7.0	6.2	DB	-1.0	39.1	1.3	1.8	-1.9	40.3	-2.7	41.4	-3.4	42.4	-4.3	43.7	(4)
			WB	-2.1	26.1	1.3	1.1	-3.1	26.9	-3.8	27.5	-4.5	28.1	-5.5	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	26.4	25.9	24.6	21.5	17.7	15.0	14.4	15.4	18.5	21.8	24.1	25.8
	DBStd	4.94	1.93	1.95	1.94	2.05	2.75	2.61	2.69	2.43	2.55	2.39	2.30	2.20
	HDD10.0	3	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	396	0	0	0	3	44	101	124	94	27	3	0	0
	CDD10.0	3982	509	445	454	346	238	152	138	168	255	366	422	490
	CDD18.3	1334	251	212	196	98	24	2	2	3	32	110	172	232
	CDH23.3	13749	2647	2166	1809	872	242	47	41	122	568	1168	1688	2378
CDH26.7	5599	1264	969	718	225	23	1	1	8	148	417	715	1111	
Wind	WSAvg	2.3	2.7	2.6	2.4	2.2	2.0	2.0	1.9	2.0	2.3	2.5	2.6	2.6
Precipitation	PrecAvg	661	95	84	54	37	39	31	26	25	27	60	86	99
	PrecMax	1052	293	284	279	198	183	128	112	109	116	241	168	344
	PrecMin	400	0	0	2	1	0	0	0	0	0	4	0	4
	PrecStd	140	64	63	53	38	39	32	30	27	30	47	45	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	37.0	35.4	32.2	28.8	26.2	25.9	27.7	32.6	34.7	36.2	37.6
		MCWB	22.4	22.0	21.4	20.1	18.1	17.5	16.3	16.4	17.6	19.5	20.9	21.6
	2%	DB	35.5	34.7	33.4	30.5	27.3	24.9	24.4	26.0	30.1	32.3	33.7	35.1
		MCWB	22.0	22.2	21.5	19.7	17.9	16.7	15.5	15.9	17.2	18.7	20.3	21.5
	5%	DB	34.0	33.1	31.9	29.2	26.1	23.6	23.2	24.7	28.4	30.6	32.2	33.4
		MCWB	21.8	22.3	21.6	19.6	17.4	16.0	14.9	14.9	16.5	18.6	20.2	21.4
	10%	DB	32.5	31.7	30.6	27.9	24.8	22.3	21.9	23.4	26.8	29.1	30.6	31.9
		MCWB	21.7	22.2	21.4	19.2	16.9	15.4	14.3	14.2	16.0	18.2	19.8	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.6	24.9	23.1	21.4	19.6	19.0	19.1	21.2	22.5	23.4	24.8
		MCDB	31.8	31.2	29.2	28.4	25.6	23.4	23.3	24.5	25.0	29.8	30.3	31.0
	2%	WB	24.5	24.7	24.0	21.9	19.8	18.0	17.6	17.8	19.5	21.1	22.5	23.8
		MCDB	30.5	30.5	29.1	27.3	24.5	22.8	21.9	22.8	25.2	27.9	29.1	30.5
	5%	WB	23.9	24.0	23.3	21.0	18.8	17.1	16.4	16.7	18.4	20.4	21.8	23.2
		MCDB	29.8	29.4	28.3	26.5	24.0	22.0	21.0	22.1	24.9	26.9	28.4	29.6
	10%	WB	23.3	23.4	22.6	20.2	17.9	16.1	15.3	15.7	17.5	19.6	21.1	22.6
		MCDB	29.0	28.9	27.8	25.7	23.1	21.0	20.2	21.2	24.5	25.9	27.9	28.8
Mean Daily Temperature Range	5% DB	MDBR	12.4	11.9	12.0	13.4	14.4	14.7	16.0	17.0	17.2	14.9	13.5	12.5
		MCDBR	15.3	14.6	14.8	15.3	15.5	15.8	17.0	18.5	19.5	17.5	15.9	14.9
		MCWBR	4.3	4.4	4.9	6.1	7.3	8.1	8.8	9.0	8.3	6.2	4.9	4.3
	5% WB	MCDBR	11.5	11.3	10.8	12.7	12.2	13.3	13.2	15.5	15.2	13.2	12.6	11.6
	MCWBR	3.9	4.0	4.4	5.7	6.3	7.4	7.4	8.3	7.4	5.2	4.3	4.0	
Clear-Sky Solar Irradiance	taub	0.382	0.372	0.352	0.332	0.311	0.302	0.295	0.309	0.345	0.372	0.382	0.390	
	taud	2.501	2.537	2.583	2.609	2.614	2.644	2.642	2.581	2.475	2.434	2.443	2.457	
	Ebn at Noon	961	957	949	926	907	896	916	937	940	945	955	956	
	Edh at Noon	115	109	99	89	81	75	77	90	109	119	122	121	
All-Sky Solar Radiation	RadAvg	6.73	6.28	5.58	4.92	4.15	3.57	4.02	4.89	5.93	6.56	6.90	6.94	
	RadStd	0.49	0.62	0.43	0.41	0.22	0.26	0.31	0.33	0.48	0.51	0.73	0.71	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page